

SmartLMS – Intelligent Learning Management Platform

Capstone Project Brief

Duration: 4 Weeks

Team Size: 10 Students

1. Business Scenario

Educational institutions use Learning Management Systems to manage courses, assessments, grades and learning activities; however, conventional LMS platforms often focus on storing and presenting information rather than actively supporting academic progress. Instructors must manually track performance, identify struggling students and communicate updates, while students may lack clear insight into their weak areas, performance trends, deadlines and expected progress. The institution requires a smart LMS that combines complete learning management with data-driven analytics, progress and risk insights, and automated workflows for reminders, alerts, reports, announcements and communication.

2. Project Goal

Build a smart Learning Management System that manages the complete learning lifecycle while providing students and instructors with performance analytics, progress insights, risk indicators, and automated academic workflows.

3. User Hierarchy

Admin

└─ Instructors

└─ Students

└─ Courses

└─ Modules & Learning Material

└─ Assignments & Quizzes

└─ Submissions & Grades

└─ Progress & Analytics

└─ Announcements

└─ Alerts & Communication

4. Core Modules

A. User & Course Management

- Admin, Instructor & Student Roles
- Courses, Enrollment & Modules
- Learning Material

B. Assessment & Progress

- Assignments & Quizzes
- Submissions & Grades
- Course/Module Progress

C. Student Analytics

- Performance Trends & Weak Areas
- Risk Indicators & Progress Projection
- Course/Module Insights

D. Instructor Analytics

- Performance Slabs & At-Risk Students
- Course/Module Performance
- Progress Reports

E. Automation & Communication

- Deadline & Inactivity Reminders
- Performance/Risk Alerts
- Reports, Announcements & Emails

F. Dashboard

- Student: Progress, grades, deadlines & insights
- Instructor: KPIs, performance, risks & progress
- Admin: Users, courses & platform statistics

6. Project Timeline

Week 1: Authentication, User Roles, Course & Module Management

Week 2: Assessments, Submissions, Grades & Progress Tracking

Week 3: Analytics, Automation, Alerts & Communication

Week 4: Testing, Deployment, Documentation & Final Presentation

7. Deliverables

- Source Code (*GitHub*)
- Documentation (*Project Report & Progress Log*)
- Demo Video (*5–10 minutes*)

8. Evaluation

- Project Planning & Teamwork – 25% (*planning, progress & teamwork*)
- Implementation – 50% (*development, AI & functionality*)
- Presentation & Viva – 25% (*demo, documentation and Q&A*)

9. Bonus Features (Optional)

- Smart study recommendations
- Calendar integration
- Certificate generation
- Advanced analytics